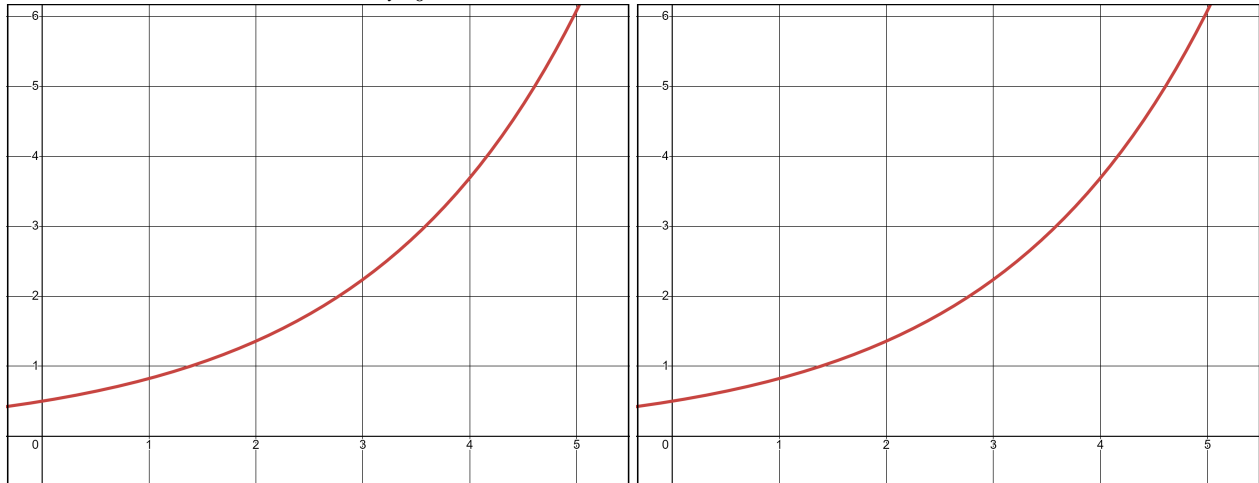


Learning target IN2 and INx, version 1

The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. Suppose that the rate of pollution (measured in tons per week) is given by the function r that is pictured below. Throughout this problem, we're interested in $\int_{t=0}^5 r(t) dt$.



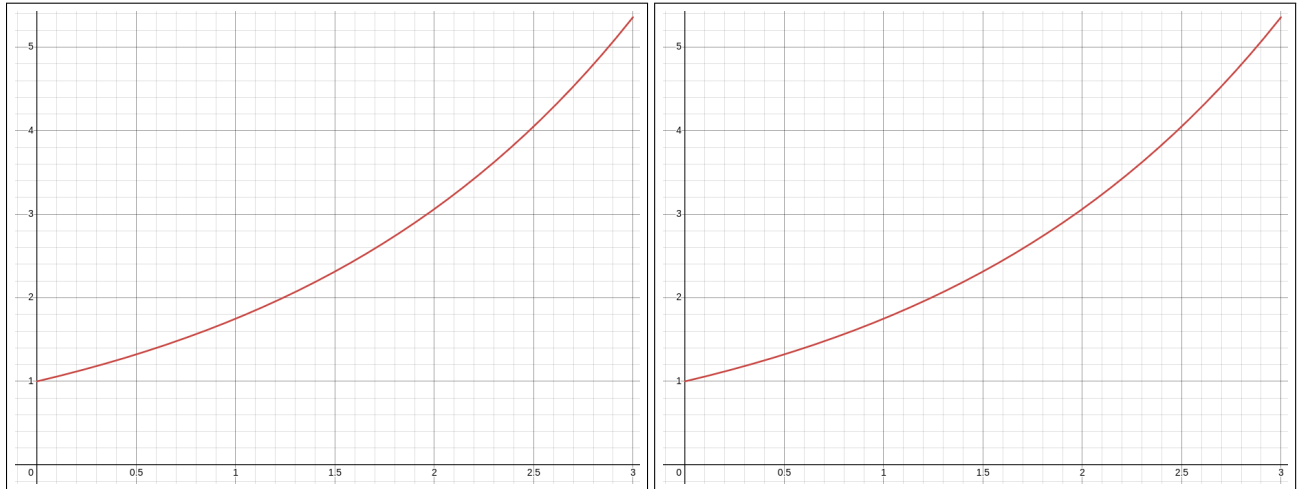
- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- If $r(t) = 0.5e^{0.5t}$, compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^5 r(t) dt$, **include units**, and explain what this number means about pollution.

Learning target IN2 and INx, version 2

A colony of bacteria in a petri dish is growing at a rate of $f(t) = 1.75^t$ million bacteria per hour. Let's figure out how much the colony grows over the course of three hours.



- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

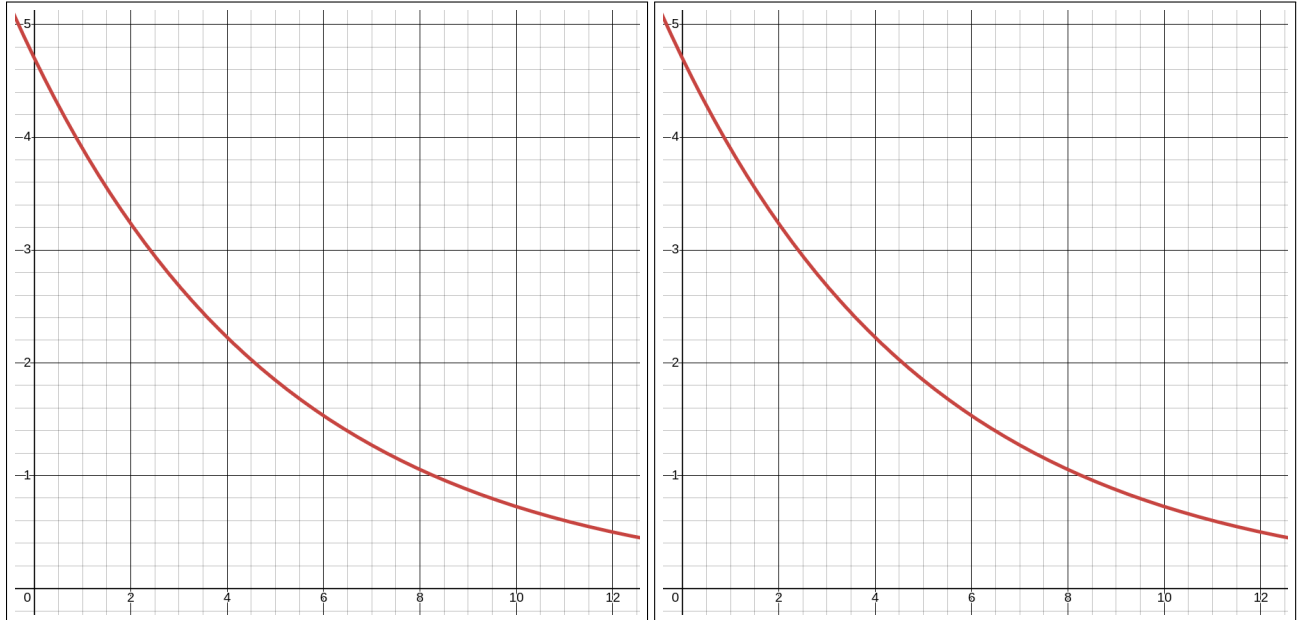
Include units on every number you write.

- Give your best guess for $\int_{t=0}^3 f(t) dt$, **include units**, and explain what this number means about the bacteria colony.

Learning target IN2 and INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$



- On the first graph above, carefully draw the *left* Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- On the second graph above, carefully draw the *right* Riemann sum with four rectangles.
- Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

- Give your best guess for $\int_{t=0}^{12} p(t) dt$, **include units**, and explain what this number means about the puppy.

Learning target INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$



- (a) On the graph above, carefully draw a Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)
- (b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}}$

the width: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}}$

the area: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{1.5cm}}_{\text{units}}$

- (c) What would the overall area $\int_{t=0}^{12} p(t) dt$ mean about the puppy? **Give units in your answer.**

Learning target INx, version 4

Suppose that an underground storage tank leaks pollutants at a rate of $r(t)$ gallons per day, where t is measured in days. A researcher studying watershed quality develops the model graphed below:

$$r(t) = 0.0069t^3 - 0.14t^2 + 11.485.$$

The researcher is interested in $\int_{t=2}^{10} r(t) dt$.



- (a) On the graph above, carefully draw a Riemann sum between $t = 2$ and $t = 10$ with four rectangles of uniform width.
(How wide should each rectangle be?)
- (b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{0.5cm}}_{\text{units}}$

the width: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{0.5cm}}_{\text{units}}$

the area: $\underbrace{\hspace{1.5cm}}_{\text{number}} \underbrace{\hspace{0.5cm}}_{\text{units}}$

- (c) What would the overall area $\int_{t=2}^{10} r(t) dt$ mean about the watershed? **Give units!**